

# SHADOWGRAM

Behavioral Graph Detection of Coordinated AI-Agent Swarms & Synthetic Persona Rings

TRACK 04: SYNTHETIC IDENTITY & KYC • TRACK 05: OPEN FRAUD

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# What is being Faked/Exploited?

## THE THREAT : CO-ORDINATED AI-ASSISTED FRAUD NETWORKS

### 1. The AI Exploit Scenario

- **Autonomous Multi-Accounting:**  
Attackers use AI agents and automation tools to operate multiple accounts in parallel.
- **Polymorphic Camouflage:**  
LLMs generate varied bios, profile text and messages, making synthetic accounts appear different and human-like.
- **Defeating Single Checks:**  
CAPTCHAs and account-level rules may fail when each synthetic account appears legitimate in isolation.

### 2. Victims & Target Sectors

- **FinTech & Micro-Lending:**  
Referral abuse, sign-up bonus abuse and coordinated account activity.
- **E-Commerce & Marketplaces:**  
Fake accounts, coordinated reviews, merchant abuse and marketplace manipulation.
- **Digital Platforms:**  
Fake engagement, phishing distribution and coordinated inauthentic activity.

### 3. Why Existing Tools Fail

- **Isolated Account Analysis:**  
Many systems examine accounts individually and may miss relationships between them.
- **Hidden Coordination:**  
Timing, navigation and behavioral similarities across accounts can remain undetected.
- **Limited Explainability:**  
A risk score such as “82% Suspicious” does not clearly show why accounts appear connected.

## CORE PROBLEM

*The fraud is not only hidden inside an account — it is hidden in the relationship between accounts.*

# How will we stop it?

## RELATIONAL BEHAVIORAL TELEMETRY & THE EXPLAINABLE "WHY" LAYER

### 1. The Relational Shift

- **Core Thesis:**

Instead of asking “**Is this account AI-generated?**”, ShadowGram asks:

“**Do these apparently independent accounts show evidence of coordinated operation?**”

- **Capturing Hidden Coordination:**

LLMs can generate varied text, but coordinated automation can still produce measurable patterns in **timing, navigation and interaction behavior**.

- **Multi-Signal Defense:**

ShadowGram combines evidence from **5 behavioral signal categories** before identifying a potentially coordinated cluster.

### 2. The 5 Behavioral Fingerprints

- **Interaction Dynamics:** Keystroke timing, dwell time and pointer movement patterns.

- **Navigation Sequences:** Repeated routes through application views, such as **Home → Search → Post**.

- **Micro-Temporal Synchronization:** Unusual similarity in the timing of actions or requests across accounts.

- **Semantic Content Similarity:** Local sentence embeddings identify similarity between apparently different messages.

- **Client Environment:**

Browser/device rendering characteristics such as **Canvas and WebGL signals**, collected with consent.

### 3. The Explainable 'Why' Layer

- **Forensic Translation:**

Converts complex graph relationships into **plain-English, auditable evidence**.

- **Inspectable Evidence:**

“Linked by 94% timing similarity, identical 4-step navigation route, and 0.89 semantic similarity.”

- **Responsible Attribution:**

The system reports “**signals consistent with coordinated operation**” rather than claiming that the accounts belong to the same person.

# Implementation

*LOCAL, DETERMINISTIC & MODULAR ENGINE ARCHITECTURE*

## 1. Systems Pipeline & Stack

- **Client Telemetry:** Captures interaction & navigation.
- **FastAPI:** Collects & processes telemetry.
- **Feature Engine:** Extracts behavioral features & embeddings.
- **NetworkX Graph:** Builds account relationship graph.
- **Louvain:** Detects coordinated clusters.
- **Dashboard:** Visualizes clusters & evidence.

## 2. Local Offline Execution

- **No External AI APIs** — Fully local processing
- **Local Embeddings** — all-MiniLM-L6-v2
- **Lightweight Graph** — Runs on a standard laptop
- **Synthetic Data** — Safe, repeatable testing

## 3. Verification & Benchmark

- **Evaluation Cohort:** 100 synthetic profiles (70 organic, 15 botnet swarms, 10 farm, 5 outliers).
- **Detection Accuracy:** 95.8% precision, 92.5% recall, 0.941 F1, <2.5% FPR.
- **Validation Flow:** Real-time Detect  
→ Cluster → Explain verification.

## CORE RUNTIME SPECIFICATIONS

**Pipeline Latency:** <150ms per batch  
**Compute Engine:** 100% Offline Local CPU

**Memory Footprint:** <150MB RAM  
**Cloud Dependency:** Zero External APIs

# Conclusion

## TRANSFORMING RISK SCORES INTO EXPLAINABLE FORENSIC INTELLIGENCE

### 1. Blast-Radius Containment

- **Detect:**

Identify hidden behavioral relationships between apparently independent accounts.

- **Explain:**

Show the signals linking accounts through an evidence-based “Why” layer.

- **Respond:**

Support targeted verification or intervention instead of relying only on blanket account bans.

### 2. Drop-In Enterprise Fit

- **Lightweight & Modular:**

Operates locally or embeds as a Trust & Safety service with minimal client overhead.

- **Integrable:**

Can be extended toward platform APIs, REST/WebSocket alerts and SIEM/SOAR workflows.

- **Privacy-Aware:**

Uses consent-based telemetry and avoids collecting passwords or unnecessary sensitive data.

### 3. Core Forensic Advantage

**Beyond:**

“82% Suspicious”

**ShadowGram:**

*“These accounts appear connected — and here is the behavioral evidence showing why.”*

## FINAL TAKEAWAY

**DETECT THE RELATIONSHIP, NOT JUST THE ACCOUNT.**

*ShadowGram turns hidden coordination into an inspectable evidence graph.*